

Entire Functions — Identity Theorem Applications & a Polynomiality Criterion

(spaced layout, matching the “Branches of $\log f$ ” style)

Background and tools

Identity (Uniqueness) Theorem. If f, g are holomorphic on a connected domain Ω and agree on a set with an accumulation point in Ω , then $f \equiv g$ on Ω .

Zeros are isolated. If h is holomorphic and not identically zero, its zeros have no accumulation point in the domain. Equivalently, if the zeros of h accumulate in the domain, then $h \equiv 0$.

Entire functions and Taylor coefficients. For entire f ,

$$f(z) = \sum_{n=0}^{\infty} c_n (z - z_0)^n, \quad c_n = \frac{f^{(n)}(z_0)}{n!}.$$

(We will use that countable unions of countable sets are countable, while $\mathbb{C}$ is uncountable.)

Problem 7. Suppose $f : \mathbb{C} \rightarrow \mathbb{C}$ is entire.

(a) If $f(1/n) = 1/n$ for all $n \in \mathbb{N}$, must $f(z) = z$ for all z ?

Yes. Let $g(z) = z$. The set $\{1/n : n \in \mathbb{N}\}$ has the accumulation point $0 \in \mathbb{C}$. Since $f(1/n) = g(1/n)$ for all n , the Identity Theorem yields $f \equiv g$ on $\mathbb{C}$. Hence $f(z) = z$ for all z .

(b) If $f(n) = n$ for all $n \in \mathbb{N}$, must $f(z) = z$ for all z ?

No. The set $\{n : n \in \mathbb{N}\}$ has no finite accumulation point in $\mathbb{C}$, so the Identity Theorem does not apply. A concrete counterexample is

$$f(z) = z + \sin(\pi z),$$

because $\sin(\pi n) = 0$ for all integers n , hence $f(n) = n$ but $f \not\equiv z$.

(c) Show that there exists some $n \in \mathbb{N}$ with $f(1/n) \neq 1/(n+1)$.

Consider $g(z) = \frac{z}{z+1}$, holomorphic on $\mathbb{C} \setminus \{-1\}$ and satisfying $g(1/n) = 1/(n+1)$ for every $n \in \mathbb{N}$. If $f(1/n) = 1/(n+1)$ held for *all* n , then f and g would agree on $\{1/n\}$, which accumulates at 0 (where both are holomorphic). By the Identity Theorem, $f \equiv g$ on $\mathbb{C} \setminus \{-1\}$. But g has a pole at -1 whereas f is entire: contradiction. Therefore there exists n with $f(1/n) \neq 1/(n+1)$.

Problem 8 (extra challenge).

Suppose $f : \mathbb{C} \rightarrow \mathbb{C}$ is entire and for every $z_0 \in \mathbb{C}$ the Taylor series

$$f(z) = \sum_{n=0}^{\infty} c_n (z - z_0)^n$$

has at least one coefficient c_n equal to 0. Prove that f is a polynomial.

Solution. For each $n \geq 0$ set $E_n = \{z \in \mathbb{C} : f^{(n)}(z) = 0\}$. Since $c_n = \frac{f^{(n)}(z_0)}{n!}$, the hypothesis says: for *every* z_0 there is some n with $z_0 \in E_n$; equivalently,

$$\mathbb{C} = \bigcup_{n=0}^{\infty} E_n.$$

If $f^{(n)} \not\equiv 0$ for every n , then each E_n is a discrete (hence countable) set by “zeros are isolated.” A countable union of countable sets is countable, contradicting that $\mathbb{C}$ is uncountable. Therefore $f^{(N)} \equiv 0$ for some N , and integrating N times shows that f is a polynomial of degree at most $N - 1$.

Problem

Identify and classify the isolated singularities of

$$\frac{\sin(2\pi z)}{z^3(2z-1)}, \quad \frac{1}{\exp(1/z) + 2}, \quad \sin^2\left(\frac{1}{z}\right),$$

and compute the residue at each isolated singularity.

Background

Classification

- **Removable:** principal part = 0.
- **Pole of order m :** finitely many negative powers, up to $(z - z_0)^{-m}$.
- **Essential:** infinitely many negative powers.

Residue (definition and formulas)

Residue at z_0 = coefficient of $(z - z_0)^{-1}$ in the Laurent series.

$$\text{Simple pole } \frac{g}{h}, \quad h(z_0) = 0, \quad h'(z_0) \neq 0 \implies \text{Res}_{z=z_0} \frac{g}{h} = \frac{g(z_0)}{h'(z_0)}.$$

$$\text{Pole of order } m \geq 1 \implies \text{Res}_{z=z_0} f = \frac{1}{(m-1)!} \lim_{z \rightarrow z_0} \frac{d^{m-1}}{dz^{m-1}} \left[(z - z_0)^m f(z) \right].$$

Basic series (for quick expansions)

$$\sin w = w - \frac{w^3}{3!} + \frac{w^5}{5!} - \dots \quad \cos w = 1 - \frac{w^2}{2!} + \frac{w^4}{4!} - \dots \quad \sin^2 w = \frac{1 - \cos(2w)}{2}.$$

Solutions

$$(i) \quad f(z) = \frac{\sin(2\pi z)}{z^3(2z-1)}$$

Singularities at $z = 0$ and $z = \frac{1}{2}$. Near 0, $\sin(2\pi z) \sim 2\pi z$, hence $f(z) \sim -\frac{2\pi}{z^2}$: **pole of order 2** at 0.

At $z = \frac{1}{2}$, $\sin(2\pi z)$ cancels $(2z-1)$, leaving a finite limit $f(1/2) = -8\pi$: **removable**.

Residues: at 0 (order 2),

$$\text{Res}_0 f = \left[\frac{d}{dz} (z^2 f(z)) \right]_{z=0} = \left[\frac{d}{dz} \frac{\sin(2\pi z)}{z(2z-1)} \right]_{z=0} = -4\pi.$$

At $z = \frac{1}{2}$: removable $\Rightarrow$ residue 0.

$$(ii) \quad f(z) = \frac{1}{\exp(1/z) + 2}$$

Solve $\exp(1/z) = -2$: with $1/z = \log 2 + i(2k+1)\pi$,

$$z_k = \frac{1}{\log 2 + i(2k+1)\pi}, \quad k \in \mathbb{Z},$$

each a **simple pole**, accumulating at 0 (so 0 is *not* an isolated singularity).

Residue at z_k :

$$\text{Res}_{z=z_k} f = \frac{1}{(\exp(1/z) + 2)'} \Big|_{z=z_k} = \frac{1}{\exp(1/z_k) \cdot (-1/z_k^2)} = \frac{z_k^2}{2}.$$

(iii) $f(z) = \sin^2\left(\frac{1}{z}\right)$

0 is an **isolated essential singularity**. Using $\sin^2 w = (1 - \cos 2w)/2$ with $w = 1/z$, the Laurent series of $\cos(2/z)$ has only even powers z^{-2n} , so the coefficient of z^{-1} is zero:

$$\operatorname{Res}_{z=0} \sin^2\left(\frac{1}{z}\right) = 0.$$

Summary.

Function	Isolated singularities	Residues
$\frac{\sin(2\pi z)}{z^3(2z-1)}$	$z = 0$ (pole order 2), $z = \frac{1}{2}$ (removable)	$\operatorname{Res}_0 = -4\pi$, $\operatorname{Res}_{1/2} = 0$
$\frac{1}{\exp(1/z)+2}$	$z_k = \frac{1}{\log 2 + i(2k+1)\pi}$ (simple poles)	$\operatorname{Res}_{z_k} = \frac{z_k^2}{2}$
$\sin^2(1/z)$	$z = 0$ (essential)	$\operatorname{Res}_0 = 0$

Maps: domains, images, zeros, poles, and level-set bands

(a) $f_1(z) = \frac{\sin(2\pi z)}{z^3(2z-1)}$

Domain. $\mathbb{C} \setminus \{0, \frac{1}{2}\}$, with a removable singularity at $z = \frac{1}{2}$ and a pole at $z = 0$.

Poles and residues.

$$\operatorname{Res}_{z=0} f_1 = \left[\frac{d}{dz} (z^2 f_1(z)) \right]_{z=0} = -4\pi, \quad z = \frac{1}{2} \text{ removable (residue 0)}.$$

Zeros. $z = n \in \mathbb{Z} \setminus \{0\}$ are simple zeros (coming from $\sin(2\pi z) = 0$).

Image/behavior. Along $\mathbb{R}$, $f_1(x) \rightarrow 0$ as $|x| \rightarrow \infty$. Near 0, $f_1(z) \sim -2\pi z^{-2}$: the mapping locally behaves like $z \mapsto 1/z^2$.

Lines and bands. The level curves $\{\Re f_1 = c\}$ and $\{\Im f_1 = b\}$ are real-analytic. Near 0 they are well approximated by the orthogonal families for $1/z^2$; bands $\{a < \Re f_1 < b\}$ narrow rapidly around the pole.

(b) $f_2(z) = \frac{1}{\exp(1/z) + 2}$

Domain. Holomorphic on $\mathbb{C} \setminus \{0\}$ except at

$$z_k = \frac{1}{\log 2 + i(2k+1)\pi}, \quad k \in \mathbb{Z},$$

where it has simple poles that accumulate at 0. Thus 0 is a non-isolated essential singularity.

Poles and residues.

$$\operatorname{Res}_{z=z_k} f_2 = \frac{1}{(\exp(1/z) + 2)'} \Big|_{z=z_k} = \frac{z_k^2}{2}.$$

Zeros. None.

Image/behavior. $f_2(z) \rightarrow \frac{1}{3}$ as $z \rightarrow \infty$. In any punctured neighborhood of 0, values are dense and every value (except possibly one) is assumed infinitely often (Picard), with the pattern organized by the lattice of simple poles z_k .

Lines and bands. Each $\{\Re f_2 = c\}$ or $\{\Im f_2 = b\}$ is an analytic curve threading between poles. Locally near z_k , the picture looks like the orthogonal families for $1/(z - z_k)$.

(c) $f_3(z) = \sin^2(1/z)$

Domain. $\mathbb{C} \setminus \{0\}$.

Poles and residues. No poles. Essential singularity at $z = 0$, and

$$\operatorname{Res}_{z=0} f_3 = 0 \quad (\text{the Laurent expansion has only even negative powers}).$$

Zeros. $z = \frac{1}{n\pi}$ for $n \in \mathbb{Z} \setminus \{0\}$, all *double* zeros, accumulating at 0.

Image/behavior. The image of any punctured neighborhood of 0 is dense in $\mathbb{C}$ and, by Great Picard, every value is assumed infinitely often (with at most one exception). In particular, 0 occurs infinitely often at the double zeros.

Lines and bands. Level sets $\{\Re f_3 = c\}$, $\{\Im f_3 = b\}$ are highly oscillatory curves clustering at 0; bands between them weave through the double-zero lattice $z = 1/(n\pi)$.

Verticals, horizontals, and bands for the three maps

(a) $f_1(z) = \frac{\sin(2\pi z)}{z^3(2z - 1)}$

Near $z = 0$: $f_1(z) = -\frac{2\pi}{z^2} + O(1)$. With $z = re^{i\theta}$,

$$\Re f_1 \approx -\frac{2\pi}{r^2} \cos(2\theta), \quad \Im f_1 \approx \frac{2\pi}{r^2} \sin(2\theta).$$

Vertical preimages $\{\Re f_1 = c\}$: $r^{-2} \cos(2\theta) \approx -c/(2\pi)$. **Horizontal preimages** $\{\Im f_1 = b\}$: $r^{-2} \sin(2\theta) \approx b/(2\pi)$. **Bands** $a < \Re f_1 < b$: regions where $-\frac{2\pi}{r^2} \cos(2\theta) \in (a, b)$; these narrow like $r \sim |c|^{-1/2}$ toward the pole.

Away from 0 the families are smooth; at $z = \frac{1}{2}$ the map extends holomorphically, so level curves pass through without singularity. As $|z| \rightarrow \infty$, $f_1(z) \rightarrow 0$, so far away the image collapses toward $w = 0$.

$$(b) \quad f_2(z) = \frac{1}{\exp(1/z) + 2}$$

Poles $z_k = \frac{1}{\log 2 + i(2k+1)\pi}$ are simple with $\text{Res}_{z_k} = \frac{z_k^2}{2}$. Near z_k ,

$$f_2(z) = \frac{\text{Res}_{z_k}}{z - z_k} + O(1) = \frac{\lambda e^{i\phi}}{z - z_k} + O(1), \quad \lambda > 0.$$

After rotation $e^{-i\phi}$ (which preserves orthogonality), we study $w = \frac{\lambda}{z - z_k}$. Writing $z - z_k = \rho e^{i\vartheta}$,

$$\Re w = \frac{\lambda}{\rho} \cos \vartheta \quad / \quad \rho = \frac{\lambda \cos \vartheta}{\rho^2}, \quad \Im w = -\frac{\lambda \sin \vartheta}{\rho^2}.$$

Vertical preimages $\{\Re f_2 = c\}$: circles/lines forming a pencil through z_k (one member is the line tangent at the pole); different c give nested curves. **Horizontal preimages** $\{\Im f_2 = b\}$: the orthogonal pencil, also through z_k . **Bands** between two levels are crescents shrinking to z_k . Because $z_k \rightarrow 0$ as $|k| \rightarrow \infty$, these pencils accumulate at 0.

As $|z| \rightarrow \infty$, $f_2(z) \rightarrow \frac{1}{3}$; far out, level curves flatten to an equipotential of a constant map.

$$(c) \quad f_3(z) = \sin^2(1/z) = \frac{1}{2}(1 - \cos(2/z))$$

Let $\zeta = 1/z$ so that $f_3(z) = \frac{1}{2}(1 - \cos(2\zeta))$. Write $\zeta = x + iy$. Then

$$\Re f_3 = \frac{1}{2}(1 - \cos 2x \cosh 2y), \quad \Im f_3 = \frac{1}{2}(\sin 2x \sinh 2y)(-1).$$

Vertical preimages $\{\Re f_3 = c\}$ correspond to $\cos 2x \cosh 2y = 1 - 2c$ in ζ -space; these are families of curves that, for moderate $|y|$, look like nearly vertical lines $x = \text{const}$ with period π . Under inversion $z = 1/\zeta$, vertical lines become circles through 0, hence the preimages in the z -plane are circles through 0.

Horizontal preimages $\{\Im f_3 = b\}$ correspond to $\sin 2x \sinh 2y = -2b$ in ζ -space; these resemble horizontal lines $y = \text{const}$ (again π -periodic in x). Their images in z -space are circles through 0, *orthogonal* to the vertical family.

Bands are inverse images of vertical/horizontal strips in ζ -space; under inversion they become narrow annular sectors around 0. The double zeros at $z = \frac{1}{n\pi}$ lie where the two families cross; all curves and bands accumulate at 0.

Problem

Classify the isolated singularities of

$$\frac{\pi}{\tan(\pi z)}, \quad \frac{z^2 - z}{1 - \sin z}, \quad \frac{\cos z - 1}{(e^z - 1)^2},$$

and compute the residue at each isolated singularity.

$$(1) \quad f_1(z) = \frac{\pi}{\tan(\pi z)} = \pi \cot(\pi z)$$

Singularities. $\tan(\pi z) = 0$ at $z \in \mathbb{Z} \Rightarrow$ simple poles at every integer.

Residues. Expanding at $z = n \in \mathbb{Z}$: $\sin(\pi z) \sim \pi(-1)^n(z - n)$, $\cos(\pi z) = (-1)^n + O((z - n)^2)$, hence $\pi \cot(\pi z) \sim \frac{1}{z - n}$. Therefore

$$\boxed{\operatorname{Res}_{z=n}(\pi \cot(\pi z)) = 1, \quad n \in \mathbb{Z}.}$$

$$(2) \quad f_2(z) = \frac{z^2 - z}{1 - \sin z}$$

Singularities. $1 - \sin z = 0 \iff z = z_k := \frac{\pi}{2} + 2\pi k, \quad k \in \mathbb{Z}$. Near z_k , $\sin z = 1 - \frac{(z - z_k)^2}{2} + O((z - z_k)^4)$, so $1 - \sin z = \frac{(z - z_k)^2}{2} + O((z - z_k)^4)$: **double poles** at each z_k .

Residues. For a double pole, $\operatorname{Res}_{z=z_k} f = \frac{d}{dz}[(z - z_k)^2 f(z)] \Big|_{z=z_k}$. Since $(z - z_k)^2/(1 - \sin z) = 2(1 + O((z - z_k)^2))$,

$$(z - z_k)^2 f_2(z) = 2(z^2 - z) + O((z - z_k)^2),$$

hence

$$\boxed{\operatorname{Res}_{z=z_k} f_2 = 2(2z_k - 1) = 4z_k - 2 \quad \left(z_k = \frac{\pi}{2} + 2\pi k\right).}$$

$$(3) \quad f_3(z) = \frac{\cos z - 1}{(e^z - 1)^2}$$

Singularities. Zeros of $e^z - 1$: $z = 2\pi ik$, $k \in \mathbb{Z}$. At $z = 0$, $\cos z - 1 = -\frac{z^2}{2} + O(z^4)$ and $(e^z - 1)^2 = z^2(1 + O(z))$, so $f_3(z) = -\frac{1}{2} + O(z)$: **removable** at 0 (residue 0).

For $z_0 = 2\pi ik$, $k \neq 0$: $(e^z - 1)^2$ has a **double pole** and the numerator is nonzero.

Residues at $z_0 = 2\pi ik$, $k \neq 0$. With $g = \cos z - 1$, $h = e^z - 1$,

$$\text{Res}_{z=z_0} \frac{g}{h^2} = \frac{g'(z_0)}{h'(z_0)^2} - \frac{g(z_0)h''(z_0)}{h'(z_0)^3}, \quad h'(z_0) = e^{z_0} = 1, \quad h''(z_0) = 1.$$

Thus

$$\boxed{\text{Res}_{z=2\pi ik} f_3 = -\sin(2\pi ik) - (\cos(2\pi ik) - 1) = -(\cosh(2\pi k) - 1 + i \sinh(2\pi k)), \quad k \neq 0.}$$

At $k = 0$ the point is removable and the residue is 0.

How vertical/horizontal lines and bands map to $w = f(z)$

$$(a) \quad f_1(z) = \pi \cot(\pi z)$$

With $z = x + iy$,

$$\cot(\pi z) = \frac{\sin(2\pi x) - i \sinh(2\pi y)}{\cosh(2\pi y) - \cos(2\pi x)}.$$

Vertical line $x = x_0$. Its image is a circle with real center:

$$\boxed{(u - \pi \cot(2\pi x_0))^2 + v^2 = \left(\frac{\pi}{\sin(2\pi x_0)} \right)^2}, \quad w = u + iv.$$

If $\sin(2\pi x_0) = 0$ (i.e. $x_0 \in \frac{1}{2}\mathbb{Z}$), this degenerates to the imaginary axis $u = 0$.

Horizontal line $y = y_0$. Parametrically

$$w(x) = \pi \frac{\sin(2\pi x) - i \sinh(2\pi y_0)}{\cosh(2\pi y_0) - \cos(2\pi x)},$$

a circle symmetric about the real axis; for $y_0 > 0$ it lies below $\mathbb{R}$ and approaches $\mathbb{R}$ as $y_0 \rightarrow \infty$ ($f_1(\mathbb{R}) \subset \mathbb{R}$ when $y_0 = 0$).

Bands. The vertical band $x \in (a, b)$ maps to the region between the two circles for $x = a$ and $x = b$. The horizontal strip $y \in (\beta, \delta)$ maps to the region between the two circles for $y = \beta$ and $y = \delta$.

$$(b) \quad f_2(z) = \frac{1}{\exp(1/z) + 2}$$

$$\text{Poles } z_k = \frac{1}{\log 2 + i(2k+1)\pi} \text{ (simple), } \text{Res}_{z_k} = \frac{z_k^2}{2}.$$

Local model near a pole z_k .

$$f_2(z) = \frac{\text{Res}_{z_k}}{z - z_k} + B_k + O(z - z_k).$$

Thus the images of straight lines through/near z_k are (to first order) arcs of the orthogonal circle/line pencils of $w = \frac{1}{z - z_k}$. **Bands** become crescent regions that shrink onto z_k . Because the poles z_k accumulate at 0, these pencils densify as $z \rightarrow 0$. Far away, $f_2(z) \rightarrow \frac{1}{3}$, so images flatten toward $w = \frac{1}{3}$.

$$(c) \quad f_3(z) = \sin^2(1/z) = \frac{1}{2}(1 - \cos(2/z))$$

Set $\zeta = 1/z = x + iy$. Then

$$\Re f_3 = \frac{1}{2}(1 - \cos 2x \cosh 2y), \quad \Im f_3 = -\frac{1}{2}(\sin 2x \sinh 2y).$$

Vertical/horizontal lines in ζ . Vertical $x = \text{const}$ and horizontal $y = \text{const}$ map under $z = 1/\zeta$ to *circles through 0* in the z -plane. Evaluating f_3 along such a circle yields a periodic oscillatory curve in w (period π in x), with higher frequency the closer the circle passes to 0.

Bands. Strips in ζ (vertical/horizontal) become annular sectors around 0 in z ; under f_3 these produce tightly interlaced images in w , reflecting the essential singularity at $z = 0$.

$$(b) \quad f_2(z) = \frac{1}{\exp(1/z) + 2}: \text{poles and residues}$$

Poles (locations and order). Solve $\exp(1/z) = -2$:

$$\frac{1}{z} = \log 2 + i(2k+1)\pi \quad \Longrightarrow \quad \boxed{z_k = \frac{1}{\log 2 + i(2k+1)\pi}}, \quad k \in \mathbb{Z}.$$

Let $g(z) = \exp(1/z) + 2$. Then

$$g'(z) = \exp(1/z) \left(-\frac{1}{z^2}\right), \quad g'(z_k) = (-2) \left(-\frac{1}{z_k^2}\right) = \frac{2}{z_k^2} \neq 0,$$

so each z_k is a *simple pole*.

Residue at z_k . For a simple pole of $1/g$, $\text{Res}_{z_k} \frac{1}{g} = \frac{1}{g'(z_k)}$. Hence

$$\boxed{\text{Res}_{z=z_k} f_2 = \frac{z_k^2}{2}}.$$

(Optional real/imag form). Write $a = \log 2$, $b = (2k+1)\pi$. Then $z_k = \frac{a - ib}{a^2 + b^2}$ and

$$\text{Res}_{z_k} f_2 = \frac{1}{2} z_k^2 = \frac{(a^2 - b^2) - i(2ab)}{2(a^2 + b^2)^2}.$$

Remark. The poles z_k accumulate at 0; therefore $z = 0$ is not an isolated singularity.

(a) $f_1(z) = \pi \cot(\pi z)$: images of vertical/horizontal lines are circles

With $z = x + iy$,

$$\cot(\pi z) = \frac{\sin(2\pi x) - i \sinh(2\pi y)}{\cosh(2\pi y) - \cos(2\pi x)},$$

so, writing $w = u + iv$,

$$u = \pi \frac{\sin(2\pi x)}{\cosh(2\pi y) - \cos(2\pi x)}, \quad v = -\pi \frac{\sinh(2\pi y)}{\cosh(2\pi y) - \cos(2\pi x)}.$$

Vertical line $x = x_0$. Eliminating y gives the exact circle

$$\boxed{(u - \pi \cot(2\pi x_0))^2 + v^2 = \left(\frac{\pi}{\sin(2\pi x_0)} \right)^2}.$$

This circle is centered at $u_0 = \pi \cot(2\pi x_0)$ on the real axis with radius $R = \pi/|\sin(2\pi x_0)|$. If $\sin(2\pi x_0) = 0$ (i.e. $x_0 \in \frac{1}{2}\mathbb{Z}$), the circle degenerates to the imaginary axis $u = 0$. Setting $u = 0$ gives $v^2 = \pi^2$, so every such circle passes through $\pm i\pi$.

Horizontal line $y = y_0$. Let $A = \cosh(2\pi y_0)$, $B = \sinh(2\pi y_0)$. Eliminating x yields

$$\boxed{\left(\frac{u}{\pi} \right)^2 + \left(\frac{v + \pi \coth(2\pi y_0)}{\pi} \right)^2 = \text{csch}^2(2\pi y_0)},$$

i.e.

$$u^2 + \left(v + \pi \coth(2\pi y_0) \right)^2 = \left(\pi \text{csch}(2\pi y_0) \right)^2.$$

The center lies at $v_0 = -\pi \coth(2\pi y_0)$ on the imaginary axis and the radius is $R = \pi \text{csch}(2\pi y_0)$. As $y_0 \rightarrow \infty$ the circle shrinks to $-i\pi$; as $y_0 \rightarrow 0$ it expands and approaches the real axis.

Mapping of lines and bands for $f(z) = \pi \cot(\pi z)$

Decomposition. With $z = x + iy$,

$$\cot(\pi z) = \frac{\sin(2\pi x) - i \sinh(2\pi y)}{\cosh(2\pi y) - \cos(2\pi x)}, \quad w = u + iv = \pi \cot(\pi z),$$

so

$$u = \pi \frac{\sin(2\pi x)}{\cosh(2\pi y) - \cos(2\pi x)}, \quad v = -\pi \frac{\sinh(2\pi y)}{\cosh(2\pi y) - \cos(2\pi x)}.$$

Image of a vertical line $x = x_0$.

$$(u - \pi \cot(2\pi x_0))^2 + v^2 = \left(\frac{\pi}{\sin(2\pi x_0)} \right)^2.$$

A circle centered at $u_0 = \pi \cot(2\pi x_0)$ on the real axis with radius $\pi/|\sin(2\pi x_0)|$. If $\sin(2\pi x_0) = 0$, the circle degenerates to the imaginary axis $u = 0$.

Image of a horizontal line $y = y_0$.

$$u^2 + (v + \pi \coth(2\pi y_0))^2 = \left(\frac{\pi}{\sinh(2\pi y_0)} \right)^2.$$

A circle centered on the imaginary axis at $-i\pi \coth(2\pi y_0)$ with radius $\pi/\sinh(2\pi y_0)$.

Bands. A vertical band $x \in (a, b)$ maps to the region between the two coaxial circles for $x = a, b$. A horizontal strip $y \in (\beta, \delta)$ maps to the region between the two circles centered on the imaginary axis for $y = \beta, \delta$.

Inverse description (from w back to z). If $(u - a)^2 + v^2 = R^2$ (center $a \in \mathbb{R}$), then it is the image of the vertical line $x = x_0$ with

$$\alpha = 2\pi x_0, \quad \cot \alpha = \frac{a}{\pi}, \quad \csc \alpha = \frac{R}{\pi} \quad (\cot^2 \alpha + 1 = \csc^2 \alpha).$$

If $u^2 + (v + \beta)^2 = \rho^2$ (center on the imaginary axis), it is the image of the horizontal line $y = y_0$ with

$$t = 2\pi y_0, \quad \coth t = \frac{\beta}{\pi}, \quad \sinh t = \frac{\pi}{\rho} \quad (\coth^2 t - 1 = \operatorname{csch}^2 t).$$

Problem

Let f be holomorphic on $\mathbb{C}$ except at finitely many singular points, all of which lie inside a positively oriented closed curve γ . Show that

$$\int_{\gamma} f(z) dz = 2\pi i \operatorname{Res}_{z=0} \left(\frac{f(1/z)}{z^2} \right).$$

Background

Residue theorem. If f is meromorphic and Γ is a positively oriented Jordan curve enclosing exactly the poles $a_1, \dots, a_m$, then

$$\int_{\Gamma} f(z) dz = 2\pi i \sum_{j=1}^m \operatorname{Res}_{z=a_j} f.$$

Residue at infinity. For a meromorphic f on the Riemann sphere,

$$\operatorname{Res}_{\infty} f := - \operatorname{Res}_{z=0} \left(\frac{f(1/z)}{z^2} \right), \quad \sum_{a \in \mathbb{C}} \operatorname{Res}_{z=a} f + \operatorname{Res}_{\infty} f = 0.$$

Equivalently,

$$\sum_{a \in \mathbb{C}} \operatorname{Res}_{z=a} f = \operatorname{Res}_{z=0} \left(\frac{f(1/z)}{z^2} \right).$$

Solution (via change of variables)

Choose $R > 0$ so that the circle $\Gamma_R := \{|z| = R\}$ encloses γ and all singularities of f . Since f is holomorphic in the annulus between γ and Γ_R , Cauchy's theorem gives

$$\int_{\gamma} f(z) dz = \int_{\Gamma_R} f(z) dz.$$

On Γ_R , set $z = \frac{1}{w}$. Then $dz = -\frac{dw}{w^2}$, and $|z| = R$ maps to $|w| = \frac{1}{R}$ with reversed orientation. Hence

$$\int_{\Gamma_R} f(z) dz = \int_{|w|=1/R} f\left(\frac{1}{w}\right) \left(-\frac{dw}{w^2}\right) (-1) = \int_{|w|=1/R} \frac{f(1/w)}{w^2} dw.$$

The integrand $\frac{f(1/w)}{w^2}$ is meromorphic in a punctured neighborhood of $w = 0$; shrinking the circle to 0 yields, by the residue theorem,

$$\int_{|w|=1/R} \frac{f(1/w)}{w^2} dw = 2\pi i \operatorname{Res}_{w=0} \left(\frac{f(1/w)}{w^2} \right).$$

Therefore,

$$\boxed{\int_{\gamma} f(z) dz = 2\pi i \operatorname{Res}_{z=0} \left(\frac{f(1/z)}{z^2} \right).}$$

Solution (residue at infinity)

Let $a_1, \dots, a_m$ be all finite singularities (inside γ). Then

$$\int_{\gamma} f(z) dz = 2\pi i \sum_{j=1}^m \operatorname{Res}_{z=a_j} f = -2\pi i \operatorname{Res}_{\infty} f = 2\pi i \operatorname{Res}_{z=0} \left(\frac{f(1/z)}{z^2} \right),$$

which is the same identity.

Residue at infinity (why this definition). Near ∞ the natural coordinate is $w = 1/z$. Then

$$f(z) dz = f\left(\frac{1}{w}\right) \left(-\frac{dw}{w^2}\right) = -\frac{f(1/w)}{w^2} dw.$$

The residue of a meromorphic 1-form is the coefficient of dw/w , hence

$$\boxed{\operatorname{Res}_{\infty} f = -\operatorname{Res}_{z=0} \left(\frac{f(1/z)}{z^2} \right)}.$$

Laurent expansion view. If $f(z) = \sum_{k=-\infty}^N a_k z^k$ for large $|z|$, then

$$\frac{f(1/z)}{z^2} = \sum_{k=-\infty}^N a_k z^{-k-2} \quad \Rightarrow \quad \operatorname{Res}_{z=0} \left(\frac{f(1/z)}{z^2} \right) = a_{-1},$$

so $\operatorname{Res}_{\infty} f = -a_{-1}$, the negative of the $1/z$ coefficient at ∞ .

Big circle integral / global residue theorem. For R large,

$$\oint_{|z|=R} f(z) dz = 2\pi i \operatorname{Res}_{z=0} \left(\frac{f(1/z)}{z^2} \right) = -2\pi i \operatorname{Res}_{\infty} f,$$

hence $\sum_{a \in \mathbb{C}} \operatorname{Res}_{z=a} f + \operatorname{Res}_{\infty} f = 0$.

Worked examples

A. $f(z) = \frac{1}{z}$

Finite residue: $\operatorname{Res}_0 f = 1$. Also

$$\frac{f(1/z)}{z^2} = \frac{1/z}{z^2} = \frac{1}{z^3} \quad \Rightarrow \quad \operatorname{Res}_{z=0} = 0, \quad \boxed{\operatorname{Res}_{\infty} f = -0 = -1}.$$

Check: $1 + (-1) = 0$, and $\oint \frac{dz}{z} = 2\pi i = -2\pi i \operatorname{Res}_{\infty} f$.

B. $f(z) = \frac{1}{z^2}$

Finite residue: $\text{Res}_0 f = 0$. Also $f(1/z)/z^2 = 1/z^4$, so $\text{Res}_{z=0} = 0$ and

$$\boxed{\text{Res}_\infty f = 0}.$$

Big circle integral is 0.

C. $f(z) = \frac{z}{(z-1)(z+2)}$ (**proper rational**)

Partial fractions: $f(z) = \frac{1}{3} \frac{1}{z-1} + \frac{2}{3} \frac{1}{z+2}$. Finite residues: $\text{Res}_1 f = \frac{1}{3}$, $\text{Res}_{-2} f = \frac{2}{3}$, sum = 1. Thus

$$\boxed{\text{Res}_\infty f = -1}, \quad \oint f dz = -2\pi i \text{Res}_\infty f = 2\pi i.$$

D. $f(z) = \frac{1}{z^2 + 1}$

At $z = \pm i$ (simple poles):

$$\text{Res}_i f = \frac{1}{2i}, \quad \text{Res}_{-i} f = -\frac{1}{2i},$$

so the sum = 0 and

$$\boxed{\text{Res}_\infty f = 0}.$$

E. $f(z) = \frac{1}{z-a}$

Finite residue: $\text{Res}_a f = 1$. Also

$$\frac{f(1/z)}{z^2} = \frac{1}{\frac{1}{z} - a} \cdot \frac{1}{z^2} = \frac{1}{z(1-az)} = \frac{1}{z} + \underbrace{a}_{\text{regular}} + \cdots,$$

so $\text{Res}_{z=0} = 1$ and

$$\boxed{\text{Res}_\infty f = -1}.$$

Check: $1 + (-1) = 0$.

F. $f(z) = P(z)$ (**polynomial**, $\deg P \geq 1$)

No finite residues. The form $f(1/z)/z^2$ contains only z^{-m} with $m \neq 1$, hence

$$\boxed{\text{Res}_\infty f = 0}, \quad \oint P(z) dz = 0.$$

G. $f(z) = \frac{1}{(z-1)(z-2)(z-3)}$ (decays like $1/z^3$)

Simple poles at 1, 2, 3. Because $f(z) = O(1/z^3)$ as $|z| \rightarrow \infty$,

$$\boxed{\operatorname{Res}_\infty f = 0}.$$

Consequently the three simple residues sum to 0 (and you can verify by partial fractions).

H. A keyhole-type integrand: $f(z) = \frac{\log(1+z)}{z^2}$

Take the principal branch with a cut along $(-\infty, -1]$. On a large circle, $f(z) = O(1/|z|^2)$, so

$$\oint_{|z|=R} f(z) dz \xrightarrow{R \rightarrow \infty} 0 \iff \operatorname{Res}_\infty f = 0.$$

Indeed,

$$\frac{f(1/z)}{z^2} = \frac{\log(1+1/z)}{z^4} = \frac{1}{z^5} - \frac{1}{2z^6} + \cdots,$$

which has no z^{-1} term; hence $\operatorname{Res}_{z=0} = 0$ and $\operatorname{Res}_\infty f = 0$. This vanishing is precisely what one uses to dismiss the big-arc contribution in the keyhole contour.

The map $f(z) = \frac{\log(1+z)}{z^2}$ (principal branch)

Domain. $\mathbb{C} \setminus ((-\infty, -1] \cup \{0\})$.

Singularities. At $z = 0$: $\log(1+z) = z - \frac{z^2}{2} + \frac{z^3}{3} - \cdots$, so

$$f(z) = \frac{1}{z} - \frac{1}{2} + \frac{z}{3} - \frac{z^2}{4} + \cdots,$$

a *simple pole* with $\boxed{\operatorname{Res}_{z=0} f = 1}$. At $z = -1$: branch point (not isolated).

At infinity. $f(z) = O((\log |z|)/|z|^2)$, hence $\oint_{|z|=R} f(z) dz \rightarrow 0$ and $\operatorname{Res}_\infty f = 0$.

Zeros. None in the domain (the only zero of $\log(1+z)$ on the principal branch is at $z = 0$, which is a pole of f).

Real-imaginary decomposition. Let $1+z = \rho e^{i\theta}$ with $\rho = |1+z|$, $\theta = \operatorname{Arg}(1+z) \in (-\pi, \pi)$, and $z = re^{i\phi}$. Then

$$u + iv = f(z) = \frac{\log(1+z)}{z^2} = r^{-2}(\ln \rho + i\theta)e^{-i2\phi},$$

hence

$$\boxed{\begin{aligned} u &= r^{-2}[(\ln \rho) \cos 2\phi + \theta \sin 2\phi], \\ v &= r^{-2}[\theta \cos 2\phi - (\ln \rho) \sin 2\phi]. \end{aligned}}$$

Level sets $\{\Re f = c\}$, $\{\Im f = b\}$ are obtained by fixing $u = c$ or $v = b$ in these formulas.

Images of straight lines ($z \rightarrow w$). For a vertical line $x = x_0$ (resp. horizontal $y = y_0$), substitute

$$r^2 = x_0^2 + y^2, \quad \phi = \arctan(y/x_0), \quad \rho^2 = (1 + x_0)^2 + y^2, \quad \theta = \arctan \frac{y}{1 + x_0}$$

(resp. with x and y_0). The resulting parametric curve $w(y)$ (resp. $w(x)$) is smooth. Near $z = 0$,

$$f(z) = \frac{1}{z} - \frac{1}{2} + O(z) \implies w + \frac{1}{2} \approx \frac{1}{z},$$

so images of vertical/horizontal lines are *approximately* the orthogonal circle/line pencils for $1/z$, shifted by $-\frac{1}{2}$ in the w -plane.

Bands. A vertical band $a < \Re w < b$ pulls back to

$$a < r^{-2}[(\ln \rho) \cos 2\phi + \theta \sin 2\phi] < b,$$

and similarly for horizontal bands using the expression for v . All bands narrow rapidly near the pole $z = 0$, as for the model map $1/z$.

Behavior near the branch point $z = -1$. With $\zeta = z + 1$, $f(z) = \frac{\log \zeta}{(\zeta - 1)^2}$. As $\zeta \rightarrow 0$, $\ln |\zeta| \rightarrow -\infty$ while $|\zeta - 1| \sim 1$; hence $|f(z)| \sim |\ln |\zeta|| \rightarrow \infty$. Small circles around -1 map to long arcs whose length grows like $|\ln |\zeta||$.

Inverse map near $w = \infty$. Solving $\log(1 + z) = wz^2$ has no closed form, but the Laurent inversion of $f(z) = \frac{1}{z} - \frac{1}{2} + \frac{z}{3} - \dots$ gives

$$\boxed{z = \frac{1}{w + \frac{1}{2}} + \frac{1}{3} \frac{1}{(w + \frac{1}{2})^3} + \frac{1}{4} \frac{1}{(w + \frac{1}{2})^4} + O\left(\frac{1}{|w|^5}\right), \quad |w| \rightarrow \infty.}$$

Thus locally $z \approx 1/(w + \frac{1}{2})$ with small rational corrections.

Problem

Use the argument principle to compute how many zeros of the polynomial

$$p(z) = z^4 + 8z^3 + 3z^2 + 8z + 3$$

lie in the right half-plane $\{\Re z > 0\}$.

Argument principle count for $p(z) = z^4 + 8z^3 + 3z^2 + 8z + 3$ in $\Re z > 0$

Let Γ_R be the positively oriented contour consisting of the segment $[-iR, iR]$ of the imaginary axis and the large semicircle $C_R = \{Re^{i\theta} : \theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]\}$ in the right half-plane. For R large, Γ_R encloses precisely the zeros of p with $\Re z > 0$. By the argument principle,

$$N_R = \frac{1}{2\pi} \Delta_{\Gamma_R} \arg p,$$

where N_R is the number of zeros (with multiplicity) inside Γ_R .

(1) Big arc C_R . As $|z| = R \rightarrow \infty$, $p(z) = z^4(1 + O(1/R))$. Hence on C_R (with $z = Re^{i\theta}$),

$$\arg p(z) = 4\theta + o(1), \quad \theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right],$$

so

$$\Delta_{C_R} \arg p \longrightarrow 4\left(\frac{\pi}{2} - \left(-\frac{\pi}{2}\right)\right) = 4\pi.$$

(2) Imaginary axis segment. Put $z = it$, $t \in [-R, R]$. Then

$$p(it) = A(t) + iB(t), \quad A(t) = t^4 - 3t^2 + 3, \quad B(t) = -8t^3 + 8t.$$

Note that

$$A(t) = (t^2 - \frac{3}{2})^2 + \frac{3}{4} > 0 \quad \text{for all } t \in \mathbb{R}.$$

Thus the curve $t \mapsto p(it)$ stays in the open right half-plane of the w -plane, never crossing the negative real axis and never encircling the origin; moreover $\arg p(it) \rightarrow 0$ as $t \rightarrow \pm\infty$. Therefore

$$\Delta_{[-iR, iR]} \arg p \longrightarrow 0.$$

(3) Count. Hence

$$\Delta_{\Gamma_R} \arg p = \Delta_{[-iR, iR]} \arg p + \Delta_{C_R} \arg p \longrightarrow 0 + 4\pi = 4\pi,$$

and by the argument principle,

$$N = \lim_{R \rightarrow \infty} N_R = \frac{1}{2\pi} \Delta_{\Gamma_R} \arg p = \frac{1}{2\pi} \cdot 4\pi = \boxed{2}.$$

Check (optional). Numerically, the roots are -7.7394 , -0.3772 , $0.0583 \pm 1.0120i$; exactly two have $\Re z > 0$.

Univalent limit or constant

Problem. Let $U \subset \mathbb{C}$ be a domain and let (f_n) be a sequence of univalent (holomorphic injective) functions $f_n : U \rightarrow \mathbb{C}$ such that $f_n \rightarrow f$ uniformly on every compact $K \Subset U$. Show that the limit f is either univalent or constant.

Background. We collect the tools we use and the exact way they are applied.

Rouché's Theorem. Let g, h be holomorphic on and inside a simple closed curve γ . If $|h(z)| < |g(z)|$ for all $z \in \gamma$, then g and $g + h$ have the same number of zeros (counted with multiplicity) in the interior of γ .

Application here. Fix $w \in \mathbb{C}$. If $f(z_0) = w$ and z_0 is the only zero of $f - w$ in a small disc D (with $|f(z) - w| > 0$ on ∂D), then the uniform convergence on ∂D gives $|f_n - f| < |f - w|$ there for all large n . Hence $f_n - w$ and $f - w$ have the same zero count in D ; in particular, $f_n(z) = w$ has at least one solution in D for all large n .

Hurwitz's Theorem. If $g_n \rightarrow g$ uniformly on compacta in a domain and each g_n is holomorphic and *non-vanishing*, then either $g \equiv 0$ or g is non-vanishing. Equivalently, zeros of g are locally the limits of zeros of g_n , counted with multiplicity.

Application here (derivative version). For univalent f_n , the inverse function theorem ensures $f'_n(z) \neq 0$ on U . If f were non-constant *and* had a critical point $f'(z_0) = 0$, then by Hurwitz applied to $g_n = f'_n$ we would get a contradiction, since g_n never vanish. Thus a non-constant limit f must satisfy $f'(z) \neq 0$ everywhere (hence be locally injective).

Two consequences for univalent maps. (i) If f is holomorphic and injective on U , then $f'(z) \neq 0$ for all $z \in U$. (ii) If $f(z_1) = f(z_2) = w$ with $z_1 \neq z_2$, then by Rouché the value w must be assumed by f_n in disjoint neighborhoods of z_1 and z_2 for all large n , contradicting injectivity of f_n .

Proof (via Rouché on fibers). Assume f is not constant. Suppose, towards a contradiction, that f is not injective: pick $z_1 \neq z_2$ in U with $f(z_1) = f(z_2) =: w$.

Choose disjoint closed discs $\overline{D_1}, \overline{D_2} \Subset U$ centered at z_1, z_2 , respectively, so small that

$$f(\partial D_j) \cap \{w\} = \emptyset \quad \text{and} \quad f - w \text{ has no zeros in } \overline{D_j} \text{ except at } z_j$$

for $j = 1, 2$. Uniform convergence on ∂D_j yields $|f_n - f| < |f - w|$ there for all $n \geq N$, hence, by Rouché, $f_n - w$ and $f - w$ have the same number of zeros in D_j . Therefore, for all $n \geq N$, the equation $f_n(z) = w$ has at least one solution in each D_j . Because $D_1 \cap D_2 = \emptyset$, this gives *two* distinct solutions, contradicting injectivity of f_n .

Thus a non-constant limit f cannot fail to be injective. Hence f is either univalent or constant. $\square$

Alternative proof (via Hurwitz on the derivatives). Assume f is not constant and $f_n \rightarrow f$ locally uniformly. Since each f_n is univalent, $f'_n(z) \neq 0$ for all z . By Hurwitz applied to (f'_n) , either $f' \equiv 0$ (which would force f to be constant, contrary to assumption) or $f'(z) \neq 0$ everywhere. Thus f is a local biholomorphism on U ; a standard covering-space/plane-domain argument then promotes local injectivity to global injectivity on U , so f is univalent.

Examples.

- *Univalent limit:* $f_n(z) = z + \frac{1}{n}$ on any domain U . Then $f_n \rightarrow f(z) = z$ (univalent).
- *Constant limit:* $f_n(z) = \frac{1}{n}z$ on any domain U . Then $f_n \rightarrow f \equiv 0$ (constant).
- *Disk automorphisms:* On $\mathbb{D}$, $f_n(z) = \frac{z}{1 - \frac{z}{n}}$ are univalent and $f_n \rightarrow f(z) = z$ uniformly on compacta.

Growth at infinity $\Rightarrow$ polynomial / rational

Problem.

- Show that if f is entire and $\lim_{z \rightarrow \infty} f(z) = \infty$, then f is a polynomial.
- Show that if f is meromorphic on $\mathbb{C}$ and $\lim_{z \rightarrow \infty} f(z) = \infty$, then f is a rational function.

Background. Let $w = 1/z$ and $g(w) = f(1/w)$.

Meromorphic at ∞ . We say f is holomorphic / meromorphic / has a pole at ∞ iff g is holomorphic / meromorphic / has a pole at 0. If g is meromorphic at 0, then for some $m \geq 1$,

$$g(w) = \sum_{k=-m}^{\infty} c_k w^k \quad (c_{-m} \neq 0),$$

and hence

$$f(z) = g(1/z) = c_{-m} z^m + \cdots + c_0 + \sum_{k \geq 1} c_k z^{-k}. \quad (*)$$

Meromorphic functions on the sphere. A function meromorphic on the Riemann sphere $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ is rational; equivalently, a meromorphic function on $\mathbb{C}$ with a non-essential singularity at ∞ is rational.

Solution to (i). Since $f(z) \rightarrow \infty$ as $z \rightarrow \infty$, $g(w) = f(1/w) \rightarrow \infty$ as $w \rightarrow 0$, so g has a pole at 0 and admits the Laurent expansion above. Using (*),

$$f(z) = c_{-m}z^m + \cdots + c_0 + \sum_{k \geq 1} c_k z^{-k}.$$

Because f is *entire*, the negative powers must vanish: $c_k = 0$ for all $k \geq 1$. Thus f is a polynomial of degree m .

Solution to (ii). Again $g(w) = f(1/w)$ has a pole at 0, so (*) holds and we can write

$$f(z) = P(z) + R(z), \quad P(z) = c_{-m}z^m + \cdots + c_0, \quad R(z) = \sum_{k \geq 1} c_k z^{-k}.$$

Here P is a polynomial and R is holomorphic for $|z| > 0$ with $R(z) \rightarrow 0$ as $z \rightarrow \infty$, so ∞ is a removable singularity of R . Therefore R extends meromorphically to the sphere and has only finitely many poles in $\mathbb{C}$. Hence R is a proper rational function (a finite sum of principal parts), and

$$f = P + R$$

is rational.

Examples.

- *Entire $\Rightarrow$ polynomial:* $f(z) = z^5 - 2z + 7$ has $f(z) \rightarrow \infty$ and is a polynomial. Any entire f with $f(z) \rightarrow \infty$ must be of this form.
- *Meromorphic $\Rightarrow$ rational:* $f(z) = \frac{z^3 + 1}{z - 2}$ satisfies $f(z) \sim z^2 \rightarrow \infty$ and is rational.
- *Non-example:* e^z is entire but $\lim_{z \rightarrow \infty} e^z$ does not exist (directional growth), so (i) does not apply.

Why “ $f'(z) \neq 0$ ” does *not* imply global injectivity

Local vs. global. If f is holomorphic on a domain U and $f'(z) \neq 0$ for all $z \in U$, then by the inverse function theorem f is a *local* biholomorphism: it is conformal and locally one-to-one on U . This does *not* force f to be globally one-to-one on U .

Counterexample. Take $f(z) = e^z$ on $\mathbb{C}$. Then $f'(z) = e^z \neq 0$ for every z , but $f(z + 2\pi i) = f(z)$, so f is not injective.

What Hurwitz really gives (in our setting). Suppose $f_n \rightarrow f$ locally uniformly on U , and each f_n is univalent. Then $f'_n(z) \neq 0$ on U . Applying *Hurwitz's theorem* to the sequence $g_n := f'_n$ yields:

$$\text{either } f' \equiv 0 \quad (\text{so } f \text{ is constant}), \quad \text{or } f'(z) \neq 0 \text{ for all } z \in U.$$

In the second case, f is a *local* biholomorphism (open and locally injective), but this alone does not ensure global injectivity.

How we get global injectivity. To rule out two distinct preimages of the same value for f , we use *Rouché's theorem on the fibers*:

- Assume $f(z_1) = f(z_2) = w$ with $z_1 \neq z_2$.
- Choose small disjoint discs $D_1, D_2 \subseteq U$ around z_1, z_2 so that $f - w$ has exactly one zero in each D_j and $|f - w| > 0$ on ∂D_j .
- Local uniform convergence gives $|f_n - f| < |f - w|$ on each ∂D_j for all large n .
- By **Rouché**, $f_n - w$ and $f - w$ have the same number of zeros in each D_j , so $f_n(z) = w$ has at least one solution in D_1 and in D_2 —contradicting the injectivity of f_n .

Therefore the nonconstant limit f is *globally* injective.

When can $f'(z) \neq 0$ imply injectivity? Additional topology is needed; for example, if U is simply connected and $f : U \rightarrow V$ is a holomorphic covering map with $f'(z) \neq 0$, then f is a conformal homeomorphism (hence injective). Without such global hypotheses, $f'(z) \neq 0$ guarantees only *local* injectivity (as e^z shows).

Holomorphic Inverse Function Theorem

Statement. Let $U \subset \mathbb{C}$ be open and let $f : U \rightarrow \mathbb{C}$ be holomorphic. If $f'(z_0) \neq 0$ for some $z_0 \in U$, then there exist neighborhoods $V \subset U$ of z_0 and $W \subset \mathbb{C}$ of $w_0 := f(z_0)$ such that

1. $f : V \rightarrow W$ is a biholomorphism (bijective, holomorphic, with holomorphic inverse),

2. the inverse $g := f^{-1} : W \rightarrow V$ is holomorphic and

$$g'(w_0) = \frac{1}{f'(z_0)}.$$

Consequences.

- If $f'(z_0) \neq 0$, then f is *locally injective* and *conformal* at z_0 .
- If $f'(z) \neq 0$ for all $z \in U$, then f is a *local biholomorphism* on U (open map, locally one-to-one).
- Global injectivity does not follow in general (e.g. $f(z) = e^z$ on $\mathbb{C}$).

Proof sketch. Because $f'(z_0) \neq 0$, the Taylor expansion at z_0 begins with a nonzero linear term:

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + o(z - z_0).$$

For w near w_0 there is exactly one z near z_0 solving $f(z) = w$; this follows from Rouché's theorem (counting solutions) or from the complex Implicit Function Theorem. The solution depends holomorphically on w , giving a holomorphic inverse g on some W . Differentiating $f(g(w)) = w$ yields $f'(g(w))g'(w) = 1$, hence $g'(w_0) = 1/f'(z_0)$. $\square$

Why “uniform on compacta” matters (counterexamples and explanations)

Set-up. Let $U \subset \mathbb{C}$ be a domain and $f_n : U \rightarrow \mathbb{C}$ be holomorphic and univalent (injective). Assume $f_n \rightarrow f$. The theorem we used says: if the convergence is *uniform on compacta*, then the limit f is either *univalent* or *constant*. Below we show, by counterexamples and mechanism, why the hypotheses “uniform” and “on compacta” are essential.

1. Pointwise convergence is too weak. One can have a sequence of univalent maps with pointwise behaviour that does not lead to a holomorphic limit, or that gives a limit without any control on injectivity nearby.

(a) No limit at all (oscillation to the boundary). On the unit disk $\mathbb{D} = \{|z| < 1\}$ consider the automorphisms

$$f_n(z) = \phi_{a_n}(z) := \frac{z - a_n}{1 - \overline{a_n}z}, \quad a_n := e^{in\theta} \left(1 - \frac{1}{n}\right), \quad \theta \in (0, 2\pi) \setminus \pi\mathbb{Q}.$$

Each f_n is a Möbius automorphism of $\mathbb{D}$, hence univalent. The parameters a_n *spin* and creep to the boundary. For a generic irrational angle θ , the sequence $(f_n(z))$ has no pointwise limit for any fixed interior point z . Thus, without some form of uniformity, there may be no holomorphic limit at all.

(b) Pointwise limit exists but gives no control. On $\mathbb{D}$ take

$$f_n(z) = \phi_{1-1/n}(z) = \frac{z - (1 - 1/n)}{1 - (1 - 1/n)z}.$$

For each fixed $z \in \mathbb{D}$, we have $f_n(z) \rightarrow -1$ (a constant) as $n \rightarrow \infty$. This shows that pointwise convergence can push images to the boundary and collapse to a constant, but pointwise control alone is insufficient to transfer local injectivity information or to apply Rouché/Hurwitz on circles.

2. Why “on compacta” is the right scale. Uniform convergence on *all of* U is often too strong (and rarely holds on unbounded domains), while *local* uniform convergence (uniform on compact subsets) is precisely what is needed to control behaviour on small circles and use zero-counting theorems.

(a) Automorphisms that “escape”, but behave nicely on compacta. The same disk automorphisms ϕ_{a_n} with $a_n \uparrow 1$ are uniformly convergent on each compact $K \Subset \mathbb{D}$ to the constant -1 , but they are *not* uniformly convergent on the whole disk. Local uniformity is exactly enough to compare f_n and the limit f on small *closed* curves contained in U .

(b) Unbounded domains: global uniformity is unrealistic. On $U = \mathbb{C}$, the translations $f_n(z) = z + n$ are entire and univalent, but they have no limit (hence not even pointwise convergence). Requiring *global* uniform convergence on $\mathbb{C}$ would force trivial limits. Uniform-on-compacta is the correct normal-family hypothesis.

3. The concrete place where uniformity is used in the proof. The proof of the “univalent limit or constant” theorem uses Rouché’s theorem on small boundary circles: for disjoint discs $D_1, D_2 \Subset U$ around a hypothetical collision $f(z_1) = f(z_2) = w$, one needs, for large n ,

$$|f_n(z) - f(z)| < |f(z) - w| \quad \text{for all } z \in \partial D_j \ (j = 1, 2).$$

This strict inequality on *every point of the boundary* is guaranteed by *uniform* convergence on the compact circles ∂D_j . If we had only pointwise convergence, then at some boundary points $|f_n - f|$ could be large, the inequality could fail on arbitrarily small arcs, and we could not conclude (via Rouché) that $f_n - w$ has a zero in each D_j . Without this, the contradiction to the injectivity of f_n collapses.

4. Summary (what each hypothesis buys you).

- **Uniform** (not just pointwise) is needed to pass zero-counts on a *curve* (Rouché/Hurwitz).
- **On compacta** is the right local version: it lets you place small circles *inside* U and control f_n uniformly there.
- Dropping either one allows sequences of univalent maps to oscillate, to run off to the boundary, or to fail to produce any holomorphic limit.